

## IM M V POPOVA, Russian Federation

WMO: 206670

Lat: 73.333N

Lon: 70.066E

Elev: 6

StdP: 101.26

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 2             | -37.2      | -35.2 | -39.8                         | 0.1 | -36.9 | -37.8 | 0.1 | -35.1 | 16.2                  | -13.8 | 14.3 | -10.8 | 3.1                   | 120  | 1.081 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 3.2                    | 14.5            | 12.5 | 12.1 | 10.5 | 10.1 | 9.0  | 12.7                | 14.3 | 10.7 | 11.9 | 9.0 | 10.0 | 4.8                  | 90   |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 11.7                            | 8.5 | 13.7 | 9.8 | 7.5 | 11.5 | 8.2 | 6.8 | 9.7  | 36.1          | 14.4 | 31.1 | 11.9 | 27.1 | 10.0 | 21.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 14.5          | 12.8 | 11.5 | DB                         | -38.3 | 18.1               | 2.9 | 4.9                                                | -40.4 | 21.6       | -42.1 | 24.5       | -43.7 | 27.3       | -45.8 | 30.8 | (4) |
| (5)               |      |      | WB                         | -38.4 | 14.8               | 2.9 | 3.4                                                | -40.5 | 17.2       | -42.1 | 19.2       | -43.7 | 21.1       | -45.8 | 23.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | -8.7   | -21.3 | -21.3 | -20.0 | -15.0 | -6.7  | 0.4   | 6.0   | 5.9   | 2.8   | -4.0  | -13.4 | -18.5 |
|                                                                                  | DBStd       | 11.93  | 8.07  | 8.87  | 7.85  | 6.17  | 4.30  | 2.39  | 3.62  | 2.22  | 2.75  | 5.34  | 6.57  | 8.37  |
|                                                                                  | HDD10.0     | 6838   | 971   | 877   | 931   | 749   | 519   | 289   | 134   | 131   | 215   | 434   | 703   | 884   |
|                                                                                  | HDD18.3     | 9864   | 1229  | 1111  | 1189  | 999   | 777   | 539   | 381   | 386   | 464   | 693   | 953   | 1142  |
|                                                                                  | CDD10.0     | 16     | 0     | 0     | 0     | 0     | 0     | 1     | 12    | 3     | 1     | 0     | 0     | 0     |
|                                                                                  | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH23.3     | 2      | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 5.6    | 5.6   | 5.4   | 5.5   | 5.4   | 5.7   | 5.5   | 5.0   | 5.6   | 5.6   | 6.0   | 5.7   | 6.0   |
| Precipitation                                                                    | PrecAvg     | 206    | 10    | 9     | 9     | 9     | 11    | 19    | 18    | 31    | 28    | 19    | 16    | 17    |
|                                                                                  | PrecMax     | 485    | 22    | 38    | 20    | 26    | 25    | 106   | 45    | 127   | 74    | 47    | 48    | 46    |
|                                                                                  | PrecMin     | 61     | 0     | 0     | 0     | 1     | 2     | 2     | 3     | 2     | 8     | 4     | 0     | 3     |
|                                                                                  | PrecStd     | 104    | 7     | 9     | 6     | 7     | 7     | 21    | 12    | 26    | 17    | 11    | 13    | 13    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | -1.8  | -1.3  | -0.9  | 0.5   | 1.5   | 13.2  | 21.7  | 15.6  | 12.4  | 4.8   | -0.1  | -1.0  |
|                                                                                  |             | MCWB   | -2.3  | -1.6  | -1.2  | 0.0   | 0.8   | 10.3  | 16.9  | 13.5  | 11.0  | 4.0   | -0.3  | -1.5  |
|                                                                                  | 2%          | DB     | -4.1  | -3.0  | -3.2  | -1.0  | 0.6   | 7.7   | 16.5  | 12.5  | 9.5   | 3.8   | -1.3  | -2.0  |
|                                                                                  |             | MCWB   | -4.5  | -3.4  | -3.6  | -1.4  | 0.1   | 6.4   | 14.0  | 10.9  | 8.8   | 3.2   | -1.6  | -2.4  |
|                                                                                  | 5%          | DB     | -6.1  | -4.9  | -5.1  | -2.8  | 0.0   | 4.7   | 13.7  | 10.8  | 7.8   | 2.7   | -2.4  | -3.4  |
|                                                                                  |             | MCWB   | -6.5  | -5.2  | -5.4  | -3.2  | -0.4  | 4.1   | 11.7  | 9.9   | 7.2   | 2.1   | -2.8  | -3.7  |
|                                                                                  | 10%         | DB     | -8.8  | -7.5  | -7.4  | -4.8  | -0.8  | 2.8   | 11.4  | 9.2   | 6.7   | 1.7   | -4.0  | -5.8  |
|                                                                                  |             | MCWB   | -9.2  | -7.8  | -7.6  | -5.2  | -1.3  | 2.3   | 9.9   | 8.3   | 6.0   | 1.2   | -4.4  | -6.1  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | -2.3  | -1.5  | -1.1  | 0.1   | 1.1   | 10.2  | 17.3  | 14.0  | 11.3  | 4.1   | -0.3  | -1.3  |
|                                                                                  |             | MCDB   | -2.0  | -1.3  | -0.9  | 0.3   | 1.4   | 13.1  | 20.5  | 15.4  | 12.1  | 4.5   | -0.3  | -1.1  |
|                                                                                  | 2%          | WB     | -4.4  | -3.3  | -3.5  | -1.4  | 0.3   | 6.6   | 14.2  | 11.3  | 8.7   | 3.2   | -1.5  | -2.4  |
|                                                                                  |             | MCDB   | -4.1  | -3.0  | -3.2  | -1.0  | 0.6   | 7.7   | 16.6  | 12.1  | 9.4   | 3.6   | -1.3  | -2.1  |
|                                                                                  | 5%          | WB     | -6.3  | -5.2  | -5.4  | -3.2  | -0.3  | 4.1   | 11.9  | 9.7   | 7.4   | 2.2   | -2.8  | -3.8  |
|                                                                                  |             | MCDB   | -5.9  | -4.9  | -5.2  | -2.9  | 0.0   | 4.6   | 13.5  | 10.6  | 7.8   | 2.5   | -2.4  | -3.5  |
|                                                                                  | 10%         | WB     | -9.1  | -7.8  | -7.7  | -5.3  | -1.2  | 2.4   | 10.0  | 8.5   | 6.1   | 1.3   | -4.4  | -6.2  |
|                                                                                  |             | MCDB   | -8.7  | -7.4  | -7.4  | -4.8  | -0.9  | 2.7   | 11.4  | 9.1   | 6.6   | 1.7   | -4.1  | -5.9  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.4   | 7.2   | 7.4   | 7.7   | 4.7   | 2.7   | 4.4   | 3.2   | 2.6   | 3.5   | 6.5   | 7.0   |
|                                                                                  |             | MCDBR  | 9.3   | 7.6   | 7.9   | 6.5   | 4.1   | 6.1   | 8.3   | 6.4   | 3.9   | 2.8   | 5.8   | 6.7   |
|                                                                                  |             | MCWBR  | 9.2   | 7.6   | 7.9   | 6.3   | 4.1   | 5.1   | 6.0   | 5.1   | 3.5   | 2.8   | 5.8   | 6.5   |
|                                                                                  | 5% WB       | MCDBR  | 9.4   | 7.7   | 7.9   | 6.5   | 4.0   | 6.1   | 8.1   | 5.6   | 3.9   | 2.7   | 5.7   | 6.8   |
|                                                                                  |             | MCWBR  | 9.2   | 7.7   | 7.9   | 6.5   | 4.0   | 5.2   | 6.2   | 4.8   | 3.6   | 2.7   | 5.8   | 6.8   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | N/A   | 0.203 | 0.277 | 0.300 | 0.324 | 0.328 | 0.362 | 0.344 | 0.305 | 0.350 | N/A   | N/A   |
|                                                                                  | taud        |        | N/A   | 1.865 | 2.001 | 2.053 | 2.076 | 2.303 | 2.405 | 2.457 | 2.421 | 2.686 | N/A   | N/A   |
|                                                                                  | Ebn at Noon |        | 0     | 316   | 634   | 788   | 823   | 844   | 794   | 759   | 654   | 317   | 0     | 0     |
|                                                                                  | Edh at Noon |        | 0     | 28    | 75    | 107   | 122   | 102   | 90    | 74    | 53    | 22    | 0     | 0     |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 0.00  | 0.15  | 1.48  | 3.87  | 5.55  | 5.61  | 4.50  | 2.68  | 1.19  | 0.25  | 0.00  | 0.00  |
|                                                                                  | RadStd      |        | 0.00  | 0.02  | 0.12  | 0.32  | 0.40  | 0.61  | 0.56  | 0.31  | 0.09  | 0.03  | 0.00  | 0.00  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                        | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page